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Research Statement

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My research program asks how institutions and markets can better account for constraints that are currently under-measured: natural capital and learning quality. I combine causal inference with economically disciplined measurement to estimate policy-relevant parameters and, where possible, connect reduced-form evidence to structural mechanisms. This agenda has two pillars. The first studies the economic value of nature in production systems and national accounting, with a focus on biodiversity and ecosystem services. The second evaluates education interventions in low- and middle-income countries to understand what improves learning at scale and why. Together, these projects are united by a common goal: producing credible evidence that helps governments and organizations allocate scarce resources more effectively.

The Economic Value of Nature

Main Work.

My core work in this agenda develops market-consistent estimates of the value of biodiversity and ecosystem services. My job market paper, *Regenerative Agriculture as Climate Mitigation* (with Jay Coggins and Nicholas Gallagher), is the central project in this line of work. The paper studies production-side climate and natural-capital margins by analyzing how land-use and farm-management choices shape mitigation outcomes, with an emphasis on the trade-offs between private production decisions and socially valuable environmental services. By quantifying these margins, the project aims to clarify when regenerative practices generate meaningful mitigation gains and how policy design can better align farm-level incentives with climate objectives. A secondary project in this stream, *Willingness to Pay for Biodiversity: Evidence from Citizen Science* (with Raahil Madhok, Matthew Braaksma, and Jovin Lasway), uses panel data on birder trips from the eBird platform in India and a mixed-logit travel-cost model to recover willingness to pay for species richness. To strengthen identification, we exploit quasi-experimental migration shocks from Siberian bird movements into India as an instrument for local biodiversity variation. This design allows us to estimate causal, spatially and temporally heterogeneous valuations of biodiversity and then map those estimates into welfare changes for local and broader populations.

Related Work.

A second stream of this agenda builds accounting frameworks that make nature visible in aggregate economic statistics. In *Global Gross Ecosystem Product (GEP)* (with NatCap TEEMs and the Chinese Academy of Sciences' RCEES), we construct comparable estimates of ecosystem-service value for 250 countries and territories in 2019 by pricing and aggregating 35 services while avoiding double counting. This work highlights both the feasibility and the remaining methodological challenges of scaling ecosystem accounting globally. In parallel, I apply the GEP framework to specific services. *Economic Contribution of Wild Pollinators: An Application of GEP* (with Subin Poudel and Lingling Liu) combines pollination sufficiency, crop-level dependence ratios, yields, and producer prices to estimate pollination-dependent production for 65 crops and identify geographic hotspots where pollinator losses would be most costly. *Ecosystem Services to Own Consumption Crop Production* (with Marta Sylla and Lijing Wang) extends these methods to subsistence systems that are often omitted from national accounts; using FAO-based interpolation and extrapolation, we estimate a 99-country panel (2004–2021) and show that nature's contribution to own-consumption crop production remains economically meaningful. Across these papers, the common contribution is a reproducible measurement strategy for integrating natural capital into policy-relevant economic aggregates.

Ongoing Project.

In *The Pollinator–Pesticide Tradeoff in Agricultural Production*, I examine a central contradiction in modern agriculture: rising demand for pollinator-dependent crops alongside input choices that can reduce wild pollinator stocks. The project studies how farmers trade off managed pollination purchases, pesticide use, and reliance on local pollination services, with the goal of identifying the market forces behind potentially inefficient input bundles. By linking these behavioral responses to production outcomes, this project aims to quantify the shadow value of common-pool natural inputs and inform policies that jointly address productivity and biodiversity conservation.

Economic Development Through Education

Main Work.

My primary education project is a prospective evaluation of Morocco's nationwide *Pioneer School Program* (with Paul Glewwe and Mridul Joshi), which combines targeted remediation and structured pedagogy. To study impacts at implementation scale, we use a matched-pair difference-in-differences design comparing 276 first-cohort schools that adopted in 2023 with 62 second-cohort schools that adopted in 2024, and we track 184 first-cohort schools into year two. We find large learning gains after two years in first-cohort schools, including Arabic (0.34 SD), French (0.91 SD), and mathematics (0.59 SD), with an overall effect of 0.59 SD; second-cohort schools show an average one-year effect of 0.46 SD. Although smaller than initial year-one estimates, these effects remain large relative to much of the evidence base from other middle- and low-income settings, providing rare evidence that high-impact pedagogical reforms can retain substantial effectiveness at scale.

Related Work.

I complement this policy-scale work with evidence on specific instructional technologies and research transparency. In *Impact Evaluation of the Darsel Math Personalized Learning Platform in Jordan* (with Caroline Krafft, Paul Glewwe, Tessa Bold, Abdulhamid Haidar, Paola Giannattasio, and Mojtaba Mahdipour; *Journal of Development Economics*, Stage 1 conditionally accepted), we evaluate whether an AI-enabled personalized math chatbot can improve grade-6 mathematics and socio-emotional outcomes in a context where enrollment is high but learning levels remain low. This project contributes to the "teaching at the right level" literature by testing a scalable personalization tool in government-school settings. In *Outsourcing Education: Experimental Evidence from Liberia* (replication of Romero, Sandefur, and Sandholtz, 2020), I reproduce and stress-test results from a major public-private-partnership experiment using a multiverse framework with 320 alternative specifications. The findings support positive effects on learning, while showing that estimated magnitudes are sensitive to some analytic choices. Together, these studies reflect my emphasis on both impact estimation and the credibility of the evidence used for policy.

Ongoing Project.

My current extension is a large synthesis of education-technology randomized evaluations using repositories such as J-PAL and Harvard Dataverse. The objective is to estimate a structural model of student learning that nests heterogeneity in EdTech design (for example, in-school versus at-home delivery, adaptive versus fixed content, and game-based versus test-based interfaces) and allows complementary or substitutable responses by teachers and parents. This framework will move beyond average treatment effects to identify the conditions under which EdTech amplifies, crowds out, or re-allocates other key educational inputs.